THE REMIX

Continuous Difference-in-Differences

From TWFE with a continuous dose to $ATT(d|d)$ and the ACRT

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Glasgow · May 2026

Causal Inference: The Remix · Yale University Press · forthcoming 2026

Continuous treatments are everywhere

When we say "DiD," we usually mean a *binary* treatment: $D_i \in \{0, 1\}$. But empirical work routinely uses DiD with a *continuous* dose:

- Minimum wages (*cents* of bite, not just "adopted")
- Abortion clinic closures (*kilometers* to the nearest clinic)
- Vaccination campaigns (*doses per 1,000*)
- Tariff cuts (*percentage-point reductions*)
- Pollution ($\mu g/m^3$)

The treatment isn't "on/off"; it has a magnitude. The question becomes "how much effect per unit of dose?" — not just "effect of treatment."

The classic praise for OLS with a continuous dose

> *The two-period regression estimator can be easily modified to allow for continuous, or at least non-binary, treatments.*

— Wooldridge (2005)

> *A second advantage of regression DiD is that it facilitates the study of policies other than those that can be described by a dummy.*

— Angrist & Pischke (2008)

The encouragement was: run the same TWFE regression, but make D a continuous variable. Easy.

The TWFE specification, with a continuous dose

The standard move is to write the same TWFE model, but let the treatment intensity vary:

$$Y_{it} = \alpha_i + \gamma_t + \beta\, D_i \cdot \text{Post}_t + \varepsilon_{it}$$

Where now:

- $D_i \in [0, \bar{d}]$ is a *dose* (kilometers, dollars, percentage points)
- Post_t is the usual post-period indicator
- β is read as “effect of one more unit of dose”

This is what almost every applied paper in the literature has used. What does β actually identify?

New work says: β^{twfe} does not have a clean causal reading

Callaway, Goodman-Bacon, and Sant'Anna (2025), *"Difference-in-Differences with a Continuous Treatment"* (**CBS**).

The core claim. Under unrestricted heterogeneity in treatment effects across doses, the TWFE coefficient $\widehat{\beta}^{\text{twfe}}$ is a *weighted average* of dose-specific effects — with weights that:

- Are *not* non-negative.
- Are *not* the dose density.
- Can place mass on *differences* between doses that the researcher never intended to compare.

We will spend the first part of this lecture unpacking these weights, on the WTO/Lu-Yu (2015) data.

Plan for the lecture

Part 1. The TWFE continuous-dose specification, and its decomposition. (Lu & Yu 2015 / China WTO accession as our running example.)

Part 2. Potential outcomes and building blocks. ATT$(d \mid d)$, the ACRT, standard vs strong parallel trends.

Part 3. Estimation: dose-specific 2×2's, overall ATT, bins, non-parametric smoothing. The `contdid` package.

We will spend more time on the decomposition than on estimation. Understanding why β^{twfe} has the structure it has is the foundation.

LESSON

Part 1:
The TWFE decomposition

Lu & Yu (2015) reported $\widehat{\beta}^{\text{twfe}} = -0.322$

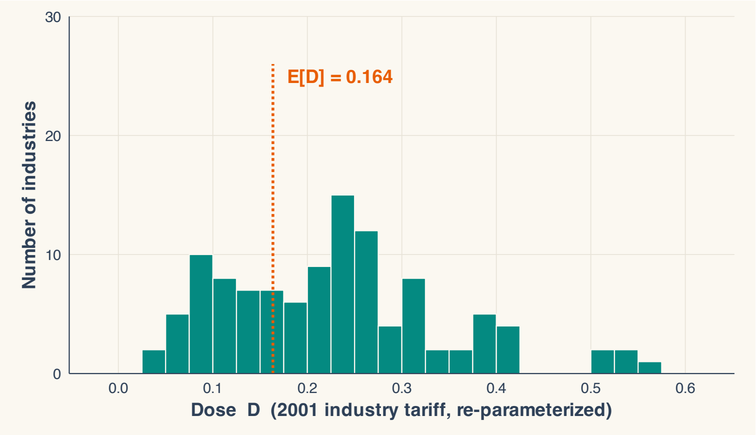
Lu & Yu (2015), *"Trade liberalization and markup dispersion: Evidence from China's WTO accession," AEJ: Applied Economics* 7(4): 221–253.

The result. A one-percentage-point cut in the import tariff facing industry i *decreased* markup dispersion by 0.322 log points. Single coefficient $\widehat{\beta}^{\text{twfe}} = -0.322$.

But $\widehat{\beta}^{\text{twfe}}$ is not the "effect of a tariff cut." It is a weighted average of dose-specific effects, with weights set by the dose distribution.

We will compute one of those weights by hand by the end of this section.

China's WTO accession: 155 industries, different tariff cuts



Distribution of D_i = tariff reduction (in percentage points) across 155 Chinese manufacturing industries. ***Mean dose*** $\approx 7\%$*. Min positive dose just above 0; max around 50%.*

One coefficient $\widehat{\beta}^{\text{twfe}}$ carries the entire dose response

The Lu & Yu specification is the canonical continuous DiD:

$$Y_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \beta^{\text{twfe}}\,(D_i \cdot \text{Post}_t) \;+\; \varepsilon_{it}$$

One coefficient. No dose-by-dose breakdown. No interaction with D_i^2 or with bins. Just β^{twfe}, the slope of Y on D in differenced units.

One coefficient, three questions:

1. Effect of any tariff cut at all?
2. Effect per percentage point of cut?
3. Effect at a particular dose, say 10%?

$\widehat{\beta}^{\text{twfe}}$ is supposed to answer all three at once. It cannot.

Imagine $\widehat{\beta}^{\text{twfe}}$ as a weighted average of dose-level effects

CBS show that, under heterogeneous dose effects:

$$\widehat{\beta}^{\text{twfe}} = \int_0^{\bar{d}} w(l) \cdot \text{ATT}(l \mid l) \, dl$$

Four different weights $w(l)$ can be motivated:

- $w^{\text{lev}}(l)$ — the "level" weight implicit in OLS
- $w^{s}(l) = l \cdot w^{\text{lev}}(l)$ — the scaled level
- $w^{\text{acrt}}(l)$ — the weight on ATT($l \mid l$) you'd want for an ACRT interpretation
- $w^{2\times 2}(l, h)$ — the weight on (l, h) pairs of doses

Each one is a real, derivable answer to "what is β^{twfe}?" They give different numbers and different shapes.

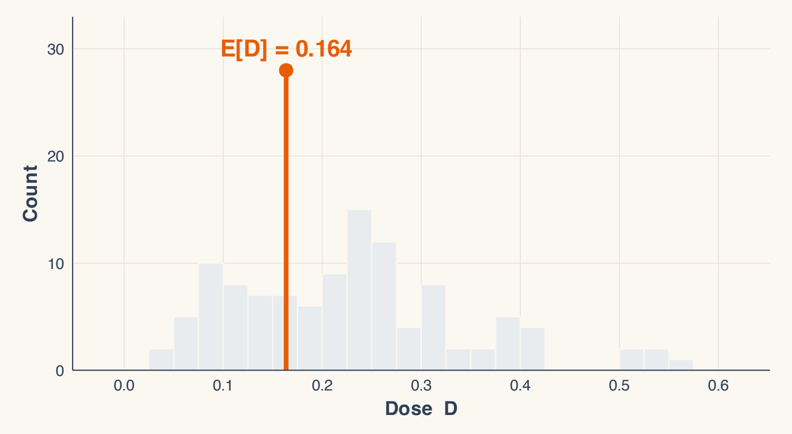
Six ingredients combine into the four weight formulas

Every weight formula is built from the same six features of the dose distribution:

1. $E[D]$ — the mean dose
2. $\text{Var}(D)$ — the variance
3. $f_D(l)$ — the density at dose l
4. $P(D \geq l)$ — the survival probability
5. $E[D \mid D \geq l]$ — the conditional mean above l
6. d_L — the minimum positive dose

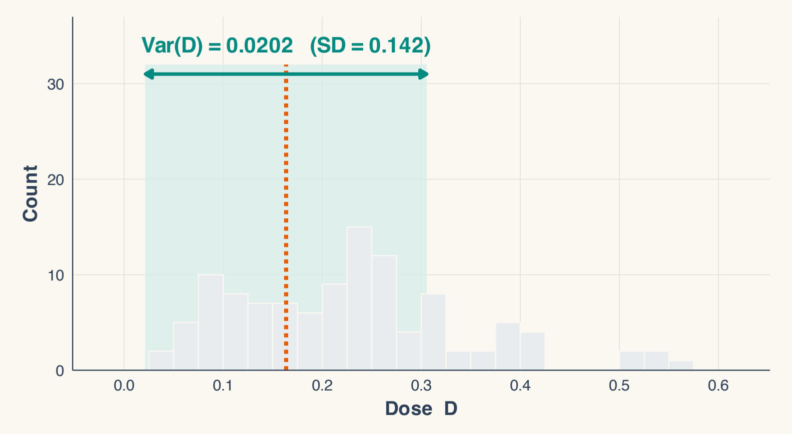
Same six numbers; four different recombinations.

Ingredient 1: $E[D]$ anchors every formula



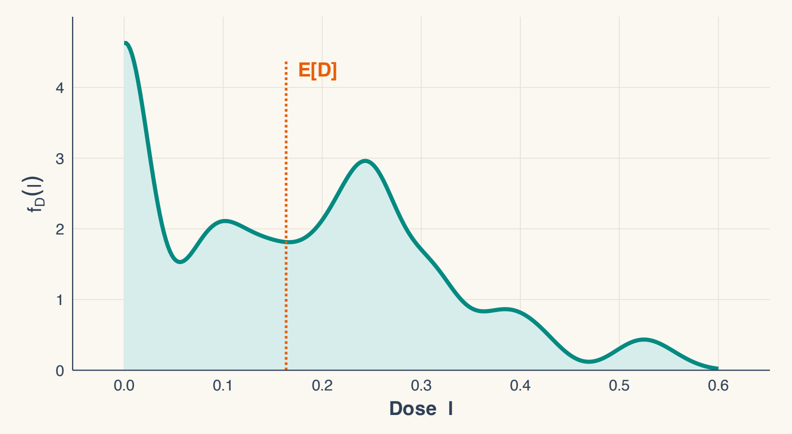
WTO mean dose $\approx 7.1\%$. The weights flip sign around this value.

Ingredient 2: Var(D) sets the scale



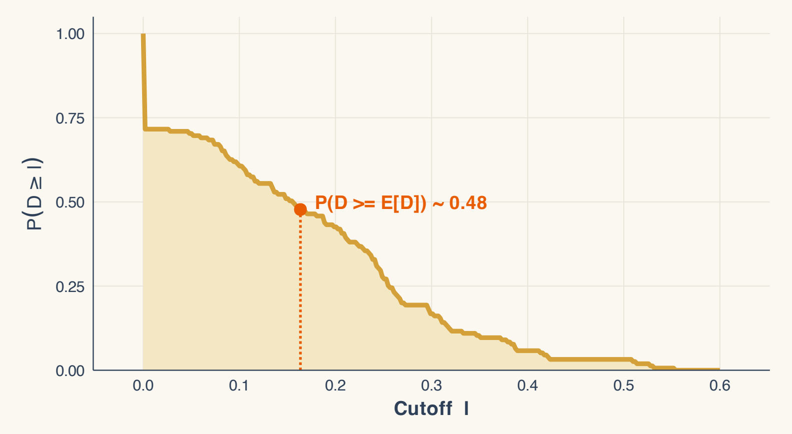
The variance shows up in the denominator of the level weight: a higher Var(D) makes the formula scale things down.

Ingredient 3: $f_D(l)$ — "where are the industries?"



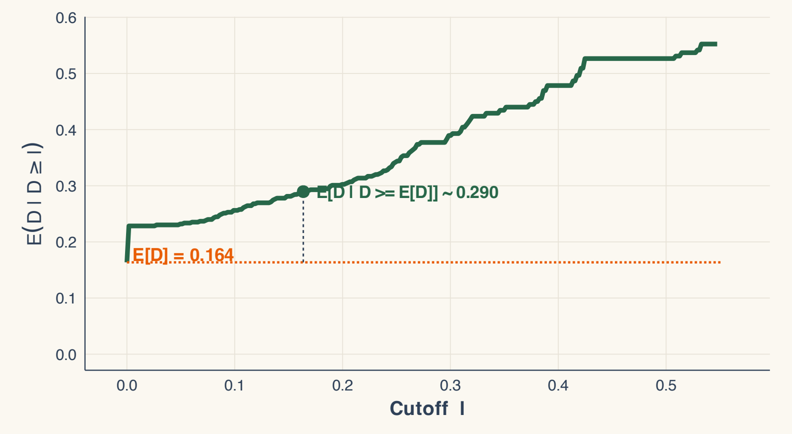
The density says where doses cluster. The ACRT weight uses f_D directly; the level weight does not.

Ingredient 4: $P(D \geq l)$ — “how many above the cutoff?”



The survival probability counts industries with dose $\geq l$. It enters the ACRT formula.

Ingredient 5: $E[D \mid D \geq l]$ tracks the upper tail



"If an industry's dose is at least l, what is its expected dose?" This conditional mean shows up in the ACRT weight.

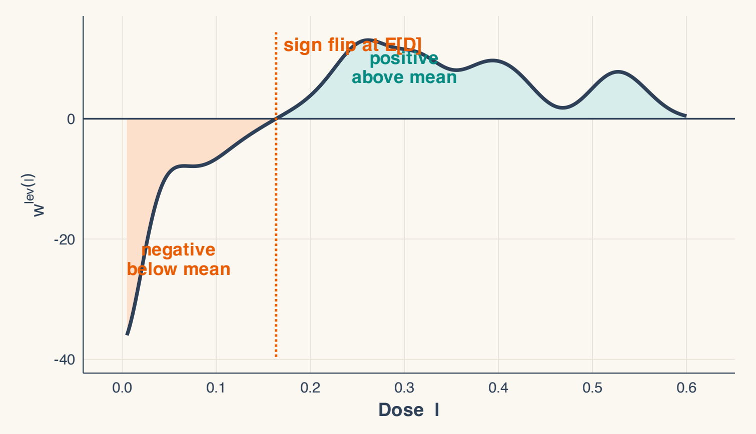
Ingredient 6: d_L, the minimum positive dose

The smallest non-zero dose in the data. In WTO it is roughly 0.5%. Doses $l < d_L$ have weight zero (no industry experiences that small a cut).

Why we name it. Several formulas integrate "from d_L to $\bar{d}$." If you fail to use d_L you can put mass on doses no one ever received — a textbook source of misinterpretation.

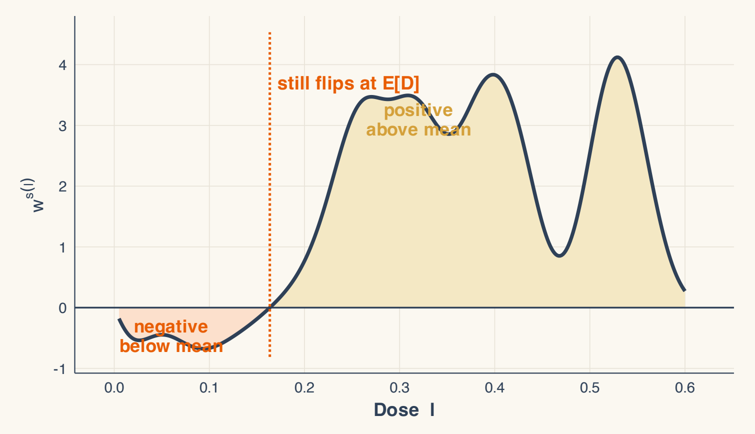
All six ingredients are computed from the cross-section of doses alone — before the outcome is even loaded.

Weight #1: the level weight $w^{\text{lev}}(l)$



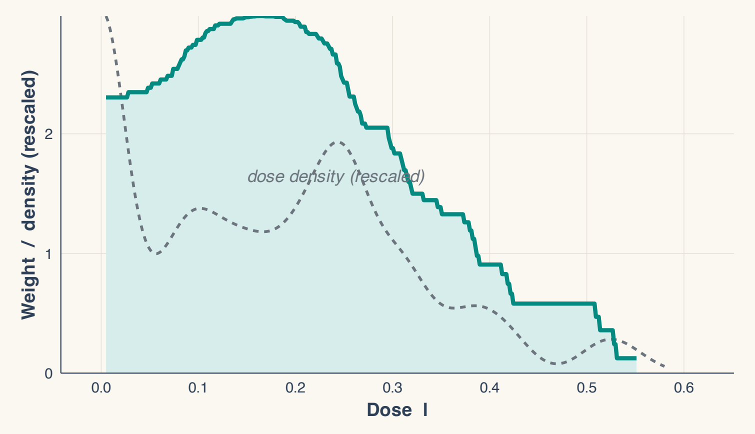
The level weight flips sign at $l = E[D]$. Below-mean doses get negative weight; above-mean doses get positive weight.

Weight #2: the scaled-level weight $w^s(l) = l \cdot w^{\text{lev}}(l)$



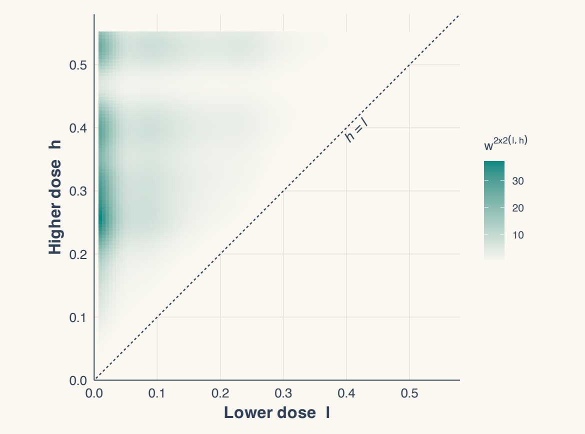
Multiplying by l pulls more weight to high doses — but the sign flip remains. It integrates to 1.

Weight #3: the ACRT weight $w^{\textsf{acrt}}(l)$



Non-negative, integrates to 1 — the only weight you'd want if you were after an average causal response. But shape differs from the density f_D.

Weight #4: the 2×2 weight $w^{2\times 2}(l, h)$



Heatmap over pairs of doses (l, h). Far-apart pairs get the most weight — $(h-l)^2$ shows up visually.

Same dose distribution, four different stories

Four overlaid weights, same distribution. $\widehat{\beta}^{twfe}$ doesn't pick one — it is each of them simultaneously, depending on which decomposition you write down.

The takeaway from the decomposition

$\widehat{\beta}^{\textbf{twfe}}$ **is well-defined.** It is a sample statistic. But:

- It is *not* the average dose effect (that would be $\int f_D(l)\,\mathrm{ATT}(l \mid l)\,dl$).
- It is *not* the ACRT (that uses w^{acrt}, not w^{lev}).
- It *is* a weighted combination, with sign-flipping weights, of dose-specific effects.

The path forward. Pick the target parameter first. Estimate that parameter. Don't read $\widehat{\beta}^{\text{twfe}}$ as "the" dose effect.

Part 2 defines those target parameters. Part 3 estimates them.

Lesson

Part 2: Potential outcomes and building blocks

ATT, but for a continuous dose

Recall the binary ATT:

$$\text{ATT} \;=\; \mathbb{E}\Big[\, Y_{it}(1) - Y_{it}(0) \mid D_i = 1,\ \text{Post}_t \,\Big]$$

With a continuous dose we need a richer object. Each unit has a potential outcome *at every possible dose*: $Y_{it}(d)$ for $d \in [0, \bar{d}]$.

Two natural parameters:

1. Effect of *being at* dose d vs being at zero. ("ATT at dose d")
2. Effect of *moving* from dose d to dose $d + \Delta$. ("causal response")

These are not the same thing.

Dosage parameters give us a multitude of ATTs

ATT for a given dose

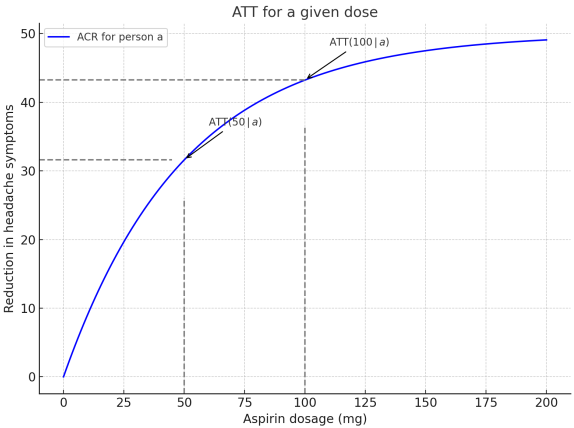
$$\text{ATT}(d \mid d) \;=\; \mathbb{E}\Big[\, Y_{it}(d) - Y_{it}(0) \mid D_i = d \,\Big]$$

Read it as: "The ATT of dose d for the group of units that actually *received* dose d, using zero dose as the counterfactual."

Example. What is the effect of being 100 km from a hospital, compared to being 0 km? Comparison is no-dose; population is the units actually 100 km away.

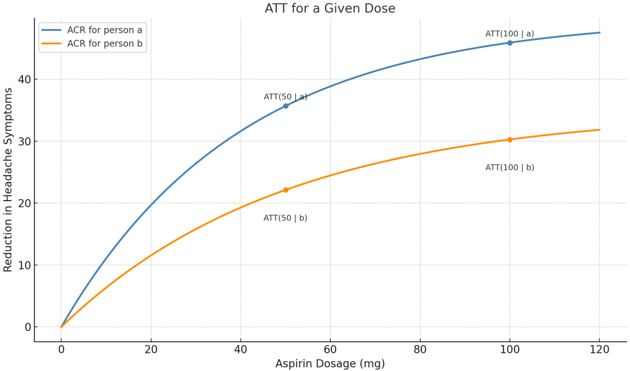
This is a function of d — one ATT per dose level. The full set $\{\text{ATT}(d \mid d)\}$ is the dose-response curve.

ATT for a given dose: the aspirin example (1)



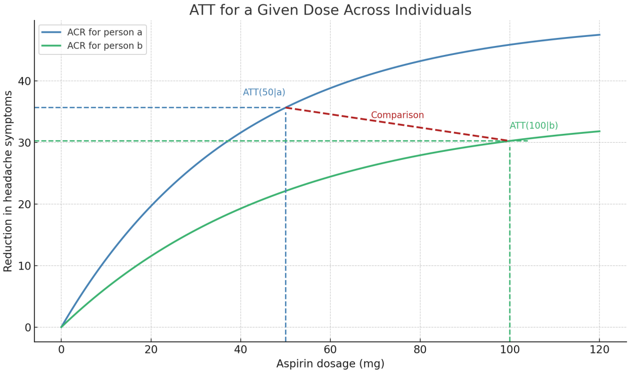
One person, one outcome (headache pain). At 50 mg, the effect is 31; at 100 mg, the effect is 44.

ATT for a given dose: the aspirin example (2)



ATT$(50 \mid 50)$ *vs ATT*$(100 \mid 100)$*: two separate ATT numbers, each comparing dose* d *to zero dose.*

Moving along the dosage is *not* the ATT



The difference between ATT(100) and ATT(50) is a slope, not an ATT. It's a marginal effect along the dose response curve.

The Average Causal Response on the Treated — ACRT

ACRT

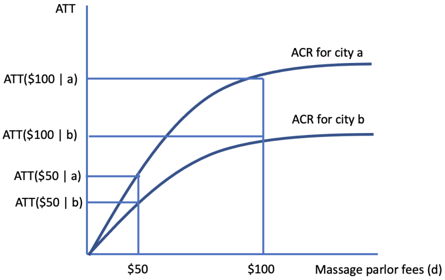
$$\text{ACRT}(d) \;=\; \frac{\partial}{\partial d}\,\mathbb{E}\Big[\,Y_{it}(d) \mid D_i = d\,\Big]$$

Read it as: "How much does the outcome change if I bump dose up by a tiny amount, evaluated at units actually receiving that dose?"

A slope, not a level. A dose-response curve: mg per mg of aspirin, km per km of distance, percentage point per percentage point of tariff cut.

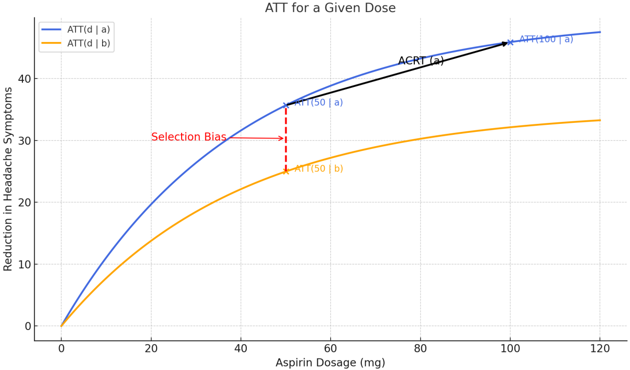
Angrist & Imbens (1995) defined the analog for IV: the Average Causal Response (ACR).

ACRT as a derivative



ACRT(d) is the slope of the ATT-dose curve at dose d — the marginal effect of one more unit of treatment.

Two different parameters, two different questions



ATT$(d \mid d)$: level effect of being at dose d. ACRT(d): marginal effect of one more unit at dose d.
Different objects, often confused.

Identification: assumptions for ATT($d \mid d$)

A1. No Anticipation. Treatment doesn't move outcomes before it begins.

$$Y_{it}(d) \;=\; Y_{it}(0) \quad \text{for all } t < g_i$$

A2. Standard Parallel Trends (PT). The trend in $Y_{it}(0)$ is the same for treated units regardless of their realized dose:

$$\mathbb{E}\Big[\Delta Y_{it}(0) \mid D_i = d\Big] \;=\; \mathbb{E}\Big[\Delta Y_{it}(0) \mid D_i = 0\Big]$$

This gives identification of ATT$(d \mid d)$ for each $d > 0$. The level-effect dose-response curve, against the no-dose counterfactual.

Standard PT is the natural extension of binary-PT to a continuous setting.

Identification: ACRT requires more — “Strong PT”

A2′. Strong Parallel Trends. The *counterfactual* trend at *every* dose is the same across treatment groups:

$$\mathbb{E}\Big[\Delta Y_{it}(d) - \Delta Y_{it}(0) \mid D_i = d'\Big] \;=\; \mathbb{E}\Big[\Delta Y_{it}(d) - \Delta Y_{it}(0) \mid D_i = d\Big]$$

In words. The treatment effect of dose d on a unit that actually received dose d equals the treatment effect of dose d on a unit that received dose d'.

This rules out selection on gains across doses. Strong PT is a substantial restriction; standard PT alone does not yield the ACRT.

Under standard PT we identify the level curve; under strong PT we identify the slope as the ACRT.

Why strong PT is stronger: a thought experiment

Suppose two industries: industry A chose a 10% tariff cut; industry B chose a 30% cut. Their dose was not assigned at random — there was a reason each chose its dose.

Standard PT says: A and B would have had the same *baseline* trend, absent any cut.

Strong PT says, in addition: *if A and B both faced a 20% cut*, they'd respond identically.

Strong PT rules out selection on dose-response gains. Exactly what an empiricist usually worries about.

Randomization of dose gives strong PT automatically; selection on the dose-response curve violates it.

Kyle's running example: the discrete multi-valued case

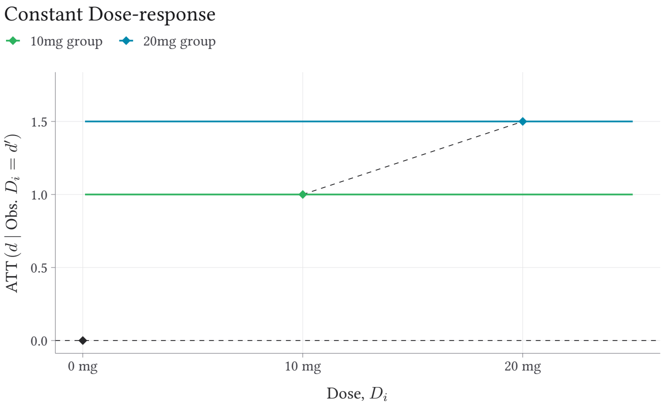
To build intuition, drop continuous for a moment and consider a *multi-valued* discrete dose: $D \in \{0, 1, 2\}$.

Setting. A doctor has patients, some untreated ($D = 0$), some receiving one pill ($D = 1$), some two pills ($D = 2$). The discrete case is the bridge between binary and continuous — all the same ideas, with finite groups.

Next slides. What dose-by-dose ATTs look like; first-differences; the DiD; what's actually identified.

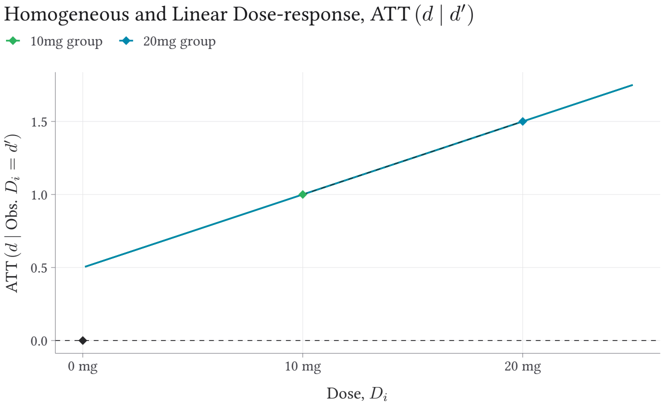
From Kyle Butts' continuous DiD slides.

Discrete: raw outcomes by dose group



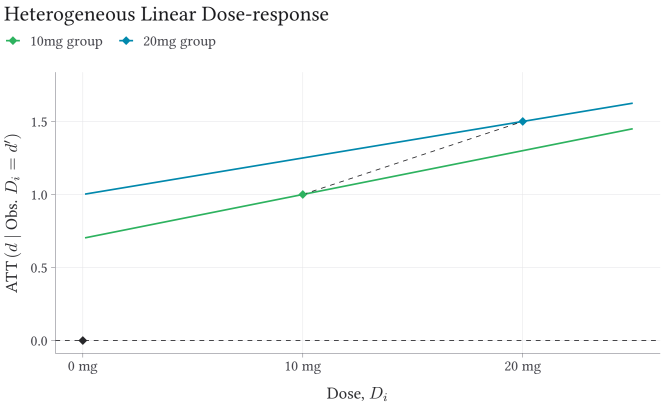
Three dose groups (0, 1, 2 pills). Their pre-period levels and trends are all parallel — this is the no-treatment-effect benchmark.

Discrete: with treatment effects (linear, homogeneous)



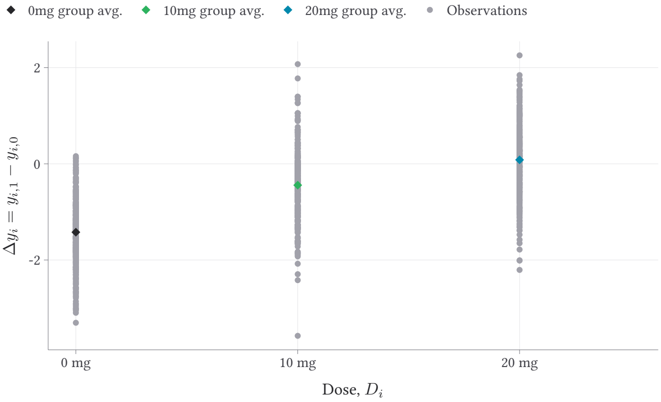
Dose-1 group lifts by δ; dose-2 group lifts by 2δ. Effects linear in dose; everyone responds the same way.

Discrete: with heterogeneity in the dose response



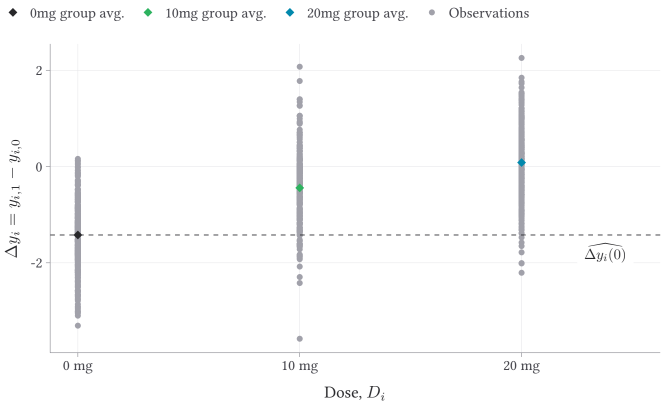
Same average response, but the dose-1 group reacts more than dose-2's first-pill response would have predicted. Heterogeneity in returns to dose.

Discrete: first differences make the comparison visible



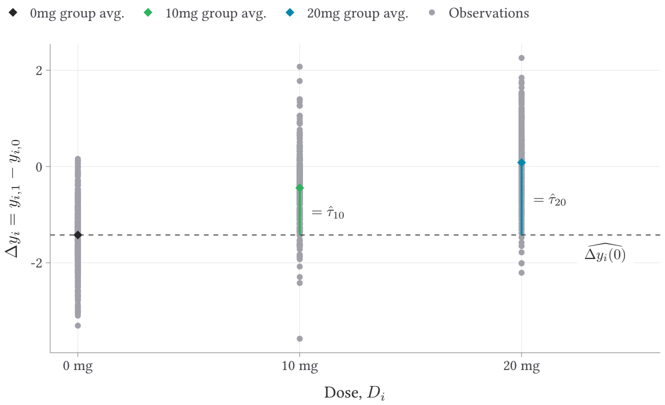
ΔY for each dose group. A clean way to read the pre-period (should be flat) vs the post-period (the treatment effect).

Discrete: the implied counterfactual



The dashed lines show what dose-1 and dose-2 would have looked like absent treatment — constructed under standard PT.

Discrete: the DiD estimates, dose by dose



Each treated dose-group's DiD versus the never-treated: $\widehat{ATT}(1 \mid 1)$ and $\widehat{ATT}(2 \mid 2)$. Two separate point estimates, one per dose.

Recap of Part 2

- Two target parameters: **ATT($d \mid d$)** (level) and **ACRT(d)** (slope).
- Standard PT identifies the level curve.
- Strong PT — ruling out selection on dose response — identifies the ACRT.
- Discrete multi-valued case gives finite-group intuition; continuous extends the same logic to a curve.

Part 3 estimates these objects, using Kyle's three options: overall ATT, bins, and non-parametric smoothing.

Lesson

Part 3: Estimation

Three estimation options for ATT($d \mid d$)

Given continuous D, three practical strategies:

1. **Overall ATT.** Average $\text{ATT}(d \mid d)$ across the treated. One number.
2. **Bins.** Partition the dose into intervals; estimate one ATT per bin. A coarse dose-response curve.
3. **Non-parametric smoothing.** A continuous dose-response function $\widehat{\text{ATT}}(\cdot)$ via kernels or splines.

Each one is a different point on the bias-variance-interpretability trade-off.

Option 1: Overall ATT — one number summary

Definition.

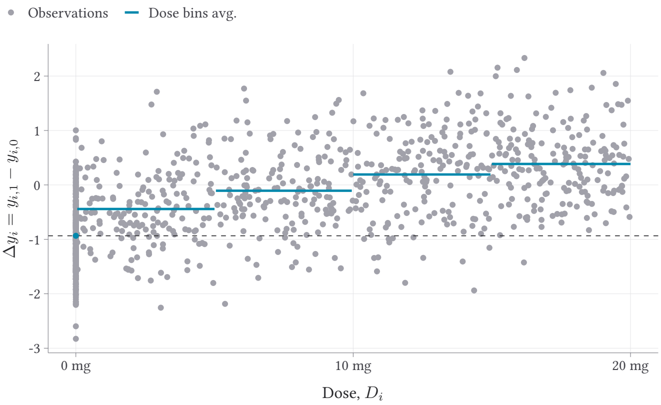
$$\overline{\text{ATT}} = \mathbb{E}\Big[\text{ATT}(D_i \mid D_i) \mid D_i > 0\Big]$$

The average of the dose-specific ATTs, taken over the treated dose distribution. The weight is the density f_D — not the level weight or any other formula.

When to use it. When the audience wants a single headline ATT. When the dose response is not the object of interest. When precision matters more than dose-by-dose detail.

This is what most empirical papers want, but it is not what TWFE delivers.

Option 2: Bins — a coarse dose-response curve



Partition dose into intervals (e.g., $[0, 5)$, $[5, 10)$, $\ldots$). Estimate one ATT per bin. Step-function dose response.

Pros and cons of bins

Pros.

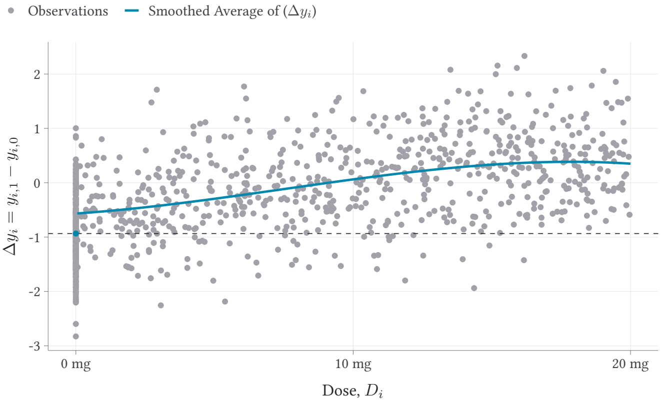
- Easy to estimate; nothing more than a series of 2×2's.
- Easy to communicate: "effect at 5–10% cut was X."
- Bins can be chosen with policy thresholds in mind.

Cons.

- Inside a bin you average over a range of doses — losing within-bin variation.
- Choice of bin width is a discretion knob.
- No standard error on the dose-response curve as a continuous object.

Often the right first cut. Rarely the final analysis.

Option 3: Non-parametric smoothing



Replace the bins with a kernel or spline. A continuous $\widehat{ATT}(\cdot)$ over the dose range, with pointwise confidence bands.

The `contdid` package

Kyle Butts's R package `contdid` implements all three options.

```
library(contdid)
result <- cont_did(
  yname = "outcome", tname = "year", idname = "unit_id",
  dname = "dose", gname = "treatment_timing",
  data = panel_data,
  target_parameter = "level", # "level" = ATT(d|d); "derivative" = ACRT
  aggregation = "dose", # or "overall"
  control_group = "notyettreated"
)
summary(result)
ggcont_did(result, type = "att")
```

`"derivative"` requires strong PT; `"level"` only standard PT.

Pros and cons of non-parametric smoothing

Pros.

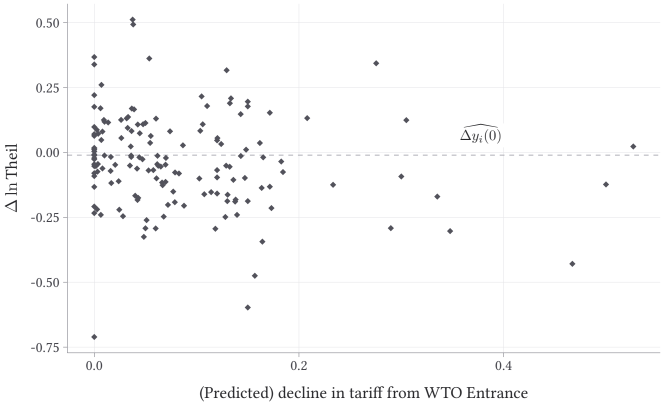
- Returns a continuous function, not a step.
- Confidence bands.
- Less ad-hoc than bin choice.

Cons.

- Bandwidth/smoothing choice is the new discretion.
- Curse of dimensionality at the tails of the dose distribution.
- Harder to communicate to a non-technical audience.

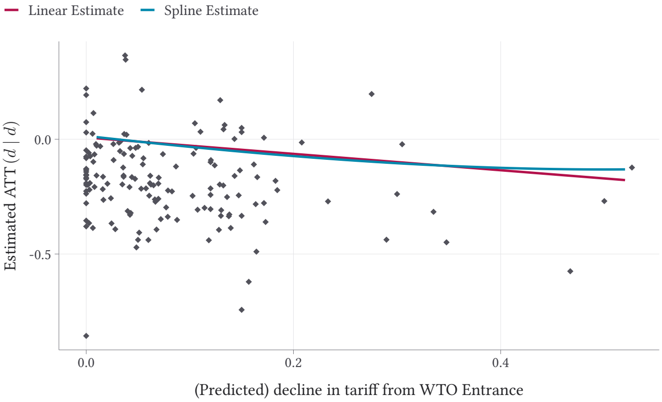
In practice: bin first, smooth second, report both.

The WTO example, done properly



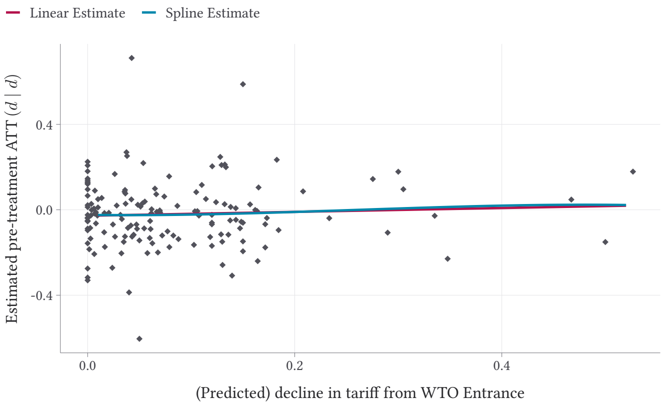
Raw outcome trajectories by tariff-cut group, before any DiD machinery. The setup we'd estimate on.

The dose-response estimates



$\widehat{ATT}(d \mid d)$ across the dose distribution. A real curve, not the single -0.322 that TWFE delivers.

Pre-trends, dose by dose



Pre-period placebo estimates. Each should be near zero under standard PT. Doses where the placebo is large are doses we shouldn't trust the post-period number on.

Closing: what to do tomorrow

Don't run TWFE with a continuous dose and report $\widehat{\beta}$ as "the dose effect." Under heterogeneity, $\widehat{\beta}^{\text{twfe}}$ is a weighted average with sign-flipping weights, scaled by features of the dose distribution that have nothing to do with the causal question.

Do:

- Pick a target parameter: **ATT($d \mid d$)** or **ACRT(d)**.
- Estimate that parameter: overall, bins, or non-parametric.
- State the identifying assumption: standard PT or strong PT.
- Report pre-trend placebos at each dose.

The continuous DiD machinery is now available — and the cost of doing it right is no longer prohibitive.